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ORIGINAL ARTICLE

Futsal task constraints promote transfer of passing skill to soccer task constraints

LUCA OPPICI[1], DEREK PANCHUK [1,2], FABIO RUBENS SERPIELLO[1], & DAMIAN FARROW[1,2]

[1]*Institute for Health and Sport, Victoria University, Melbourne, Australia &* [2]*Movement Science, Australian Institute of Sport (AIS), Canberra, Australia*

Abstract
This study investigated how learning a passing skill with futsal or soccer task constraints influenced transfer to a new task. Futsal ($n = 24$, 13.6 ± 1.2 years old, 7.0 ± 1.6 years of experience) and soccer ($n = 24$, 13.6 ± 1.2 years old, 6.8 ± 1.2 years of experience) players performed two 5v5 + goalkeeper modified games – a futsal-like task (small playing area with the futsal ball) and soccer-like task (large playing area with the soccer ball). Participants' passing accuracy and their orientation of attention were assessed during the two tasks. The futsal group improved their passing accuracy (ES = 0.75 ± 0.61) from the futsal-like to the soccer-like task, and they were more accurate than soccer players (ES = 2.98 ± 2.96). Conversely, the soccer group's passing accuracy remained stable across the two tasks (ES = 0.10 ± 0.52) and it was similar to the futsal group in the futsal-like task (ES = 0.58 ± 1.93). This indicates a higher magnitude of transfer (and adaptability) from performing passes in a small playing area with short time to act – futsal task constraints – to a larger playing area with longer time – soccer task constraints – than vice-versa. Furthermore, the futsal group showed a higher adaptation of attention orientation to the affordances that emerged with the soccer task constraints, which is suggested to be one of the main mechanisms promoting skill transfer. These results encourage soccer practitioners to introduce futsal task constraints to fast-track players' ability to functionally adapt perception–action coupling.

Keywords: *Skill adaptability, attention orientation, passing skill, soccer, futsal, football*

Highlights
- Learning the passing skill with futsal task constraints promoted higher transfer than practicing with soccer task constraints.
- Futsal players' adaptability of attention orientation underpinned their superior skill transfer.
- Futsal task constraints can enhance passing skill in soccer.

Introduction

The constraints-led perspective contends that an individual's behaviour in goal-directed activities emerges from the self-organisation of interacting organismic, environmental and task constraints (Newell, 1986). The constraints interaction shapes the emergence of opportunities for action, called affordances (Newell, 1991). With practice, individuals become perceptually attuned to environmental information that specifies affordances and develop the ability to organise coordination patterns functional to the achievement of the task goal (Araújo & Davids, 2011; Davids, Button, & Bennett, 2008). Skill learning involves a relatively permanent change in behaviour over time as a result of goal-directed practice with a set of constraints (Kelso & Zanone, 2002; Newell, 1996; Zanone & Kelso, 1997). Transfer of learning is the process of adapting that behaviour to a new set of constraints (Newell, 1996; Rosalie & Müller, 2012). Transfer is evaluated on performance achievement, not on an 'idealised' movement reference, as different behaviours can lead to successful performance (Travassos, Araujo, & Davids, 2018). Transfer is positive when individuals successfully adapt their behaviour and improve their performance; negative when performance

Correspondence: Luca Oppici, Institute for Health and Sport, Victoria University, PO Box 14428, Melbourne VIC 8001, Australia. E-mail: luca.oppici@live.vu.edu.au

decreases, or neutral when performance is not affected (Carroll, Riek, & Carson, 2001). Despite transfer of learning being recognised as a central issue in motor skill performance (Adams, 1987; Shea, Kovacs, & Panzer, 2011), the practice conditions that foster the transfer process remain an open question (Broadbent, Causer, Williams, & Ford, 2015; Pacheco & Newell, 2015). It has been suggested that practicing with certain task constraints, for example, equipment and rules, guides the development of an adaptive behaviour that, in turn, promotes skill transfer (Davids et al., 2008); however, the relationship between task constraints and transfer of learning in sport is relatively unexplored.

The similarity of information that guides action is what promotes skill transfer between tasks (Pinder, Davids, Renshaw, & Araújo, 2011; Snapp-Childs, Wilson, & Bingham, 2015). Research in sport has shown that similarity between domains promoted transfer of perceptual-motor skills from basketball to darts (Rienhoff et al., 2013) and from baseball to cricket (Moore & Müller, 2013). However, since the interaction of task constraints shapes information (Newell, 1991), domain-specific task constraints are likely to influence skill transfer. Research in bimanual oscillatory finger movement showed how task constraints during practice influenced transfer. Participants who practised a bimanual oscillatory finger movement at a 90-degree relative phase demonstrated transfer of the coordination pattern primarily to an untrained, 270-degree symmetrical phase (Zanone & Kelso, 1997). Given this research used a relatively simple task, it is unclear how practicing a sport skill with different task constraints influences transfer. Here, we investigated this issue evaluating transfer of passing skill between futsal and soccer task constraints.

Futsal is the 5-a-side indoor form of football, whereas association football (also called soccer) is the 11-a-side outdoor form of football, both of which are officially regulated by the Fédération Internationale de Football Association (FIFA). Passing is the main skill performed in both domains (Mohammed, Shafizadeh, & Platt, 2014; Rampinini, Impellizzeri, Castagna, Coutts, & Wisløff, 2009) and is a complex perceptual-motor skill that involves the reception of the ball and a pass towards a teammate. Throughout the action, players are required to simultaneously perceive the surrounding environment and to execute the coordinated action to successfully deliver the ball to a teammate. Spatial and temporal information of attacker–defender interactions shapes the emergence of passing affordances during games (Travassos, Araújo, Davids, Vilar, et al., 2012). Specifically, relative angles and distances between the ball carrier and teammates, and the ball carrier and a teammate's nearest defender constrain the passing action. A player's ability to adapt his perceptual behaviour and action to the continuous changes of these relationships underpins successful passing performance (Corrêa, Vilar, Davids, & Renshaw, 2014a, 2014b; Travassos, Araújo, Davids, Esteves, & Fernandes, 2012). As such, players are required to attune their attention simultaneously towards this key environmental information to decide on who to pass the ball to, and towards the approaching ball to guide the execution of reception and delivery of the ball (Oppici, Panchuk, Serpiello, & Farrow, 2017).

The relative similarity of information that guides the passing action in both sports is expected to promote transfer of passing skill to the other domain. However, passing in the two domains is practised with different task constraints (see Table I for more details) that influence perception–action coupling, and, in turn, affect the transfer process. The smaller individual playing area in futsal likely leads to more frequent and rapid changes of interplayer relationships (i.e. relative distance and angles) than soccer, which could fast-track players' ability to explore the information that specifies passing affordances. This might be facilitated by the futsal ball's lower coefficient of restitution (Peacock, Garofolini, Oppici, Serpiello, & Ball, 2017) and flat surface that simplify ball control. It has been shown that futsal players explored different information (i.e. ball and players) more frequently and oriented their attention to players' movement for longer than

Table I. Domain-specific task constraints are presented.

Task constraints	Domain Futsal	Soccer
Number of players	5 vs. 5	11 vs. 11
Pitch size	40 × 22.5 m	100 × 65 m
Individual playing area	90 m^2/player	295 m^2/players
Rules	Offside	No offside
Ball circumference	62.5–63.5 cm	68.5–69.5 cm
Ball coefficient of restitution (Peacock, Garofolini, Oppici, Serpiello and Ball)	0.51	0.60

soccer players when performing passes during games (Oppici et al., 2017). Perceptual-motor exploration during practice has been shown to facilitate skill adaptability and, in turn, transfer of skills (Bennett, Button, Kingsbury, & Davids, 1999). Therefore, it is likely that (the magnitude of) transfer will be higher from futsal to soccer task constraints than vice-versa. However, it is unclear how practicing the passing skill with futsal or soccer task constraints influences transfer.

The aims of this study were (i) to determine how learning the passing skill with domain-specific task constraints influences transfer to new task constraints and (ii) to explore the perceptual skills underpinning transfer. Passing performance and attention orientation were assessed in young, skilled (i.e. with more than 1000 h of domain-specific practice) futsal and soccer players during modified games. Participants performed a game with their own-domain task constraints and a game with the other-domain task constraints to evaluate transfer. Although a certain degree of transfer in both groups is expected, given the relative similarity of information that guides passing between the learning and transfer tasks, we predicted a higher magnitude of transfer for futsal players, due to the more demanding environment in which passing has been developed. Therefore, it is expected that futsal players would improve their passing performance from futsal to soccer task constraints, while soccer players' passing performance would remain stable across the two modified games.

Methods

Participants

Forty-eight ($n = 48$) skilled, young, male players were recruited for the experiment. They were divided into two groups based on their domain of expertise: soccer players ($n = 24$, 13.6 ± 1.2 years old, 6.8 ± 1.2 years of experience) and futsal players ($n = 24$, 13.6 ± 1.2 years old, 7.0 ± 1.6 years of experience). The years of experience refer to structured team practice in their domain. Participants completed a questionnaire on their training history, which included the number of training years at the club level, an average amount of training months per season, number of training sessions per week and session duration. Participants had 1220 ± 216 and 1260 ± 288 hours of domain-specific structured practice in soccer and futsal, respectively. Only players that had never practised the other sport at the club level were included in the sample. Both groups were recruited from elite clubs and had a similar weekly training schedule that included four 90-minute training sessions and one competitive game. Only outfield players were sampled from the squads. Technical issues during data collection (e.g. the scene camera moved during the game and unreliable video quality) resulted in a final sample size of 17 futsal players and 20 soccer players.

Prior to the study, participants were fully informed of the risks involved in participating in the experiment and their parents (or guardian) provided written consent for them to participate. The study was approved by the research team's University Ethics Committee.

Experimental task and procedure

The experimental task was a 5v5 plus goalkeeper (5v5 + GK) game. Two tasks were designed to sample the task constraints typical of passing (i.e. playing space, ball and surface) in the two sports. A large-area game representing soccer (SOC) was performed outdoor with an FIFA quality-approved soccer ball, on a synthetic-grass pitch of 24 m × 36 m corresponding to a playing area of 86 m^2/player, which is representative of the common playing space during soccer game (Fradua et al., 2013). A small-area game representing futsal (FUT) was performed indoor with an FIFA quality-approved futsal ball, on a wooden pitch of 24 m × 15 m corresponding to a playing area of 36 m^2/player, which is the most common density in futsal matches (unpublished observations). The two tasks did not maintain the same number of players and pitch dimension of their relative sport. However, inter-player interactions were sampled from the two sports using playing area as a key constraint; thus, these tasks can be considered representative of short passing skill in futsal and soccer.

An external camera (GoPro 3+, California, USA) was placed in one corner of the pitch to record the task, allowing the off-line evaluation of pass outcome. The scene camera of a mobile eye tracking system (Mobile Eye, Applied Sciences Laboratories, Bedford, MA, USA) was used to collect participants' attention orientation.

Participants performed three sessions in total; a familiarisation session in the other-domain task, which consisted of a shortened version of the experimental session, and two experimental sessions, their own-domain task first and, then, the other-domain task. The sessions were interspersed by 48 hours. Each experimental session composed of six games that were 5 minutes in duration with 5-minute breaks between games.

After a standardised, 10-minute warm-up, each group was randomly divided into two teams of six players plus a goalkeeper. In both tasks, futsal players played against futsal players and soccer players played against soccer players. Before each task, two participants were fitted with the Mobile Eye (e.g. participants A and B wore the camera in game 1, then participants C and D wore the camera in game 2, and so on). The task started with the investigator bouncing the ball on the ground. The bounce was used as a reference to synchronise the eye-tracker scene camera and the external camera. One of the investigators umpired the games. At the end of the session, each player wore the scene camera during one game and participated in five games, resting the game prior to wearing the scene camera.

Data analysis

Pass accuracy. Participants' pass accuracy was evaluated as the main performance variable. A pass was considered accurate when the ball reached a teammate (i.e. the teammate made contact with the ball before any opponent), consistent with previous research (Serpiello, Cox, Oppici, Hopkins, & Varley, 2017). Pass accuracy was evaluated using a commercially available video analysis software (Sports Code, SportsTec by Hudl, Australia).

Attention orientation. Attention orientation and game dynamics variables are briefly presented as further details are published elsewhere (Oppici et al., 2017). Participants' attention orientation was classified into two areas of interest using the centre of the image from the scene camera as a reference, consistent with Pluijms et al. (2016) previously validated methods. The orientation of attention was classified either as ball-directed, whenever the ball was in the centre of the image, or as player-directed, when the ball was not in the centre. The footage from the scene camera and the external camera, both recorded at 30 Hz, were synchronised using Sports Code to couple the orientation of attention with specific events, consistent with the Vision-in-Action procedure (Vickers, 1996). The game was divided into three phases: reception (the time from a teammate's pass to participant's first touch), control (from participant's first touch to pass release) and team phase (time of participant's team ball possession minus the previous two phases). Three attention-orientation variables were determined: relative attention-orientation time in the two areas of interest (AT), attention-orientation switches between the two areas (AS) and last-attention location (AL). AT refers to the relative percentage of time the attention was oriented on the two areas of interest. AS was calculated as the number of times participants alternated their attention between the two areas per second. AL refers to the location toward which the head was oriented when the foot made contact with the ball, to control or to kick the ball. These dependent variables were evaluated frame-by-frame using Sports Code. AT and AS were evaluated, separately, in each phase, whereas AL was examined only in the reception and control phases.

Passes where the receiving pass (from the teammate) was bouncing and the participant's body was oriented towards the sideline (with less than 4 players in front of him) were excluded from the analysis to limit the influence of external factors. Furthermore, only passes that included the control phase were considered (i.e. direct passes were excluded). This resulted in a total of 731 passes analysed, 369 in FUT (169 futsal group, 200 soccer group) and 362 in SOC (181 in both groups).

Game dynamics. Three game-related variables were evaluated to quantify the context in which the passes were performed: technical intensity (i.e. number of passes per minute), individual playing area surrounding the ball carrier (IPA) and reception time. IPA refers to the playing area the player was in when performing the pass. The pitch was divided into squares of known dimensions and the number of players inside the square the participant was occupying when performing the pass was counted. The dimension of the square was divided by the number of players to calculate IPA. Reception time refers to the time it took the ball to travel from a teammate to the participant wearing the Mobile Eye.

Coding reliability. Five percent of the trials were randomly selected and independently coded by two coders, and then re-coded a week later by the primary coder for inter- and intra-rater reliability. Cohen's kappa was above 0.9 in all dependent variables in both intra- and inter-rater reliability, representing perfect agreement (Landis & Koch, 1977).

Statistical analysis

Pass accuracy and attention-orientation dependent variables were analysed separately using generalised linear mixed modelling with repeated measures (Proc Glimmix in Version 3.6 of Statistical Analysis System Studio, SAS Institute, Cary, NC), with group (soccer, futsal) and task (SOC, FUT) as fixed factors and participants as a random factor. Allowance was made for overdispersion. Specifically, a logistic regression was computed for pass accuracy, AT and AL, being binary- dependent variables, whereas Poisson regression was computed to analyse AS as data were expressed as count per unit of time (switches/second). Furthermore, pass accuracy was adjusted for game dynamics, having compared two different groups' outcome in the same task. The between-subject standard deviation for the standardisation of the effect sizes (ES) was calculated using the pure observed between-subject variance and the overdispersed sampling variance. The game dynamics variables were analysed separately using an independent *t*-test for the two tasks.

Significance was set at $p < .05$ for all the analyses and the magnitude of changes was assessed using ES with 90% confidence intervals defined as follows: <0.2 trivial, 0.2–0.6 small, 0.6–1.2 moderate, 1.2–2.0 large, >2.0 very large (Hopkins, 2010).

Results

Pass accuracy

The mean data of the two groups are presented in Table II. There was a positive significant task effect in the futsal group ($p = .046$, ES = 0.75 ± 0.61) and no significant task effect in the soccer group ($p = .74$, ES = 0.10 ± 0.52). Futsal players improved their passing accuracy from FUT to SOC, while pass accuracy of soccer players remained stable across the two tasks. Furthermore, there was a very large, near-significant group effect in pass accuracy in SOC ($p = .09$, ES = 2.98 ± 2.96) and no group effect in FUT ($p = .61$, ES = 0.58 ± 1.93). Futsal players performed more accurate passes than soccer players in SOC, while there was no group difference in FUT.

Attention orientation

The mean data for attention-orientation variables is presented in Table II.

Reception phase. In the reception phase, there was a significant group effect in AL in FUT (p<.01, ES = 1.35 ± 0.58), while there was a small, near-significant group effect in AL in SOC ($p = .07$, ES = 0.58 ± 0.53). Furthermore, there was a significant task effect in both futsal ($p < .01$, ES = 1.41 ± 0.52) and soccer groups ($p = .02$, ES = 0.62 ± 0.45) in AL (Figure 1). Prior to first touch, futsal players oriented their attention towards other players more times than soccer players in FUT, while this difference was smaller in SOC; both groups oriented their attention towards other players more times in FUT than SOC. No group or task effect was found in AS and AT.

Control phase. In the control phase, there was a task effect in the futsal group in AL ($p = .048$, ES = 0.57 ± 0.47) and in AT ($p = .01$, ES = 0.73 ± 0.45).

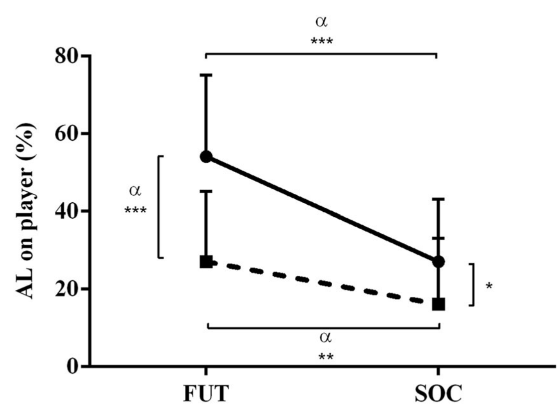

Figure 1. AL on player-directed areas in the reception phase. The futsal group is presented with the solid line, while the soccer group with the dashed line. *α*: statistically significant, effect sizes: *small, **moderate, ***large.

Table II. Pass accuracy (in percentage), mean relative attention-orientation time on player-directed area (AT, in percentage), attention-orientation switches (AS, in switches/second) and last-attention orientation on ball-directed area (AL, in percentage) ± SD of the two groups across the three different phases in the two tasks.

		Within-group comparison						Between-groups comparison	
		Futsal group			Soccer group			*p* value	
Phase	Variable	FUT	SOC	*p* value	SOC	FUT	*p* value	FUT	SOC
	Pass accuracy	75 ± 12	84 ± 15	.046**	80 ± 12	80 ± 12	.7	.6*	0.09****
Reception	AT	17 ± 12	13 ± 9	.2*	14 ± 10	15 ± 10	.9	.4*	0.8
	AS	1.0 ± 0.5	0.9 ± 0.5	.4*	0.6 ± 0.3	0.8 ± 0.4	.4*	.2*	0.3*
	AL	54 ± 21	27 ± 16	<.01***	16 ± 17	27 ± 18	.02**	<.01***	0.07*
Control	AT	50 ± 19	39 ± 15	.01**	34 ± 18	40 ± 19	.1*	.1*	0.5
	AS	0.9 ± 0.3	0.8 ± 0.3	.1*	0.9 ± 0.3	0.9 ± .04	.9	.3*	0.4*
	AL	20 ± 23	9 ± 13	.05*	14 ± 21	18 ± 20	.5	.7	0.4*
Team	AT	13 ± 5	15 ± 7	.2*	22 ± 10	17 ± 10	<.01*	.2*	0.04**
	AS	0.3 ± 0.1	0.3 ± 0.1	.8	0.4 ± 0.1	0.4 ± 0.1	.8	.01**	<0.01**

Significance was set at $p < .05$. Effect sizes: *small, **moderate, ***large and ****very large.

Futsal players oriented their attention towards the ball more times in SOC than FUT prior to performing the pass, and for a longer period throughout the phase. No group effect or task effect in the soccer group was found in AT and AL, and no group or task effect in AS.

Team phase. In the team phase, there was a group effect in SOC ($p = .04$, ES $= 0.73 \pm 0.58$) and a task effect in the soccer group ($p < .01$, ES $= 0.54 \pm 0.29$) in AT. Soccer players oriented their attention towards other players for longer than futsal players in SOC, and for longer in SOC than FUT. No group effect was found in FUT, and no task effect in the futsal group. Furthermore, there was a group effect in FUT ($p = .01$, ES $= 0.87 \pm 0.54$) and in SOC ($p < .01$, ES $= 0.99 \pm 0.60$) in AS. Soccer players switched their attention between ball and players more frequently than futsal players in both tasks. No task effect was found in both groups.

Game dynamics

There was a significant group effect in technical intensity in FUT ($p < .01$, ES 3.25 ± 0.48) and in SOC ($p < .01$, ES 4.34 ± 0.48). The futsal group performed a higher number of passes per minute than the soccer group in both tasks. There was a group effect in IPA in FUT ($p < .01$, ES 0.91 ± 0.54), while there was no effect in SOC. The futsal group had a lower individual playing area than the soccer group in FUT. No group effect was found in reception time in both tasks.

Discussion

This study examined how learning a passing skill with futsal or soccer task constraints influenced transfer to the other domain's task constraints. Despite expecting a certain degree of transfer in both futsal and soccer players, we hypothesised a higher magnitude of transfer from futsal practice to the soccer task constraints due to the effect of practicing passes with the more demanding futsal task constraints (i.e. smaller individual playing area, shorter reception time and higher technical intensity). The hypothesis was confirmed as the futsal group significantly improved their passing accuracy from FUT to SOC, while the passing accuracy of the soccer group remained stable. This indicated positive transfer of passing skill from futsal practice to soccer task constraints, and neutral transfer for the soccer group. Furthermore, the results showed that futsal players were more accurate with higher game intensity than soccer players in SOC, while soccer players performed passes with similar accuracy but lower game intensity than futsal players in FUT.

Transfer is the process of adapting a learned behaviour to new constraints (Newell, 1996; Rosalie & Müller, 2012), and these results indicated that practicing the passing skill in a small space with a short time to act (futsal task constraints) facilitated a higher degree of transfer to a larger space with a relatively longer time to act (soccer task constraints) than vice-versa. Futsal players' adaptation of perceptual behaviour to the passing affordances that emerged in SOC might have facilitated skill transfer. The participants had practised passing exclusively within their own domain (i.e. ball, playing surface and individual playing area) for more than 1000 h, and thus the perceptual behaviour of futsal players in FUT and soccer players in SOC could be viewed as a functional attunement to passing affordances in FUT and SOC, respectively. For example, information about the ball during the reception and control phases seemed to be critical with the soccer task constraints (Table II). This might be important to ensure a good impact with the ball at reception and delivery. Futsal players significantly changed their orientation of attention during these two phases from FUT to SOC, increasing AL and AT toward the ball, which resulted in a similar behaviour to soccer players (Figure 1). The futsal group's higher passing accuracy in SOC suggests that futsal players not only adapted their attention orientation according to the emergence of passing affordances in SOC but they were also better able to perceive information that specified affordances and acted accordingly, resulting in a more functional coupling of passing actions. On the other hand, soccer players' orientation of attention remained stable across the two tasks (except for AL in the reception that slightly changed) which might explain the group's stable passing performance.

This study's novel findings have theoretical and practical implications. While previous research highlighted the influence of futsal and soccer task constraints on the development of perceptual skill underpinning passing (Oppici et al., 2017), this study shows how learning a passing skill with futsal and soccer task constraints affects transfer. Researchers have suggested that practicing with a certain set of task constraints develops an adaptive behaviour promoting transfer of learning (Araújo, Davids, Bennett, Button, & Chapman, 2004; Davids et al., 2008). Skill learning in team sports entails exploring perception–action relations, discovering and stabilising functional solutions and exploiting affordances and degrees of freedom (Araújo, Davids, Chow, & Passos, 2009). Task constraints that encourage exploration of information during practice can fast-track the stabilisation of functional movement solutions in a transfer task (Pacheco & Newell, 2015). For example, practicing a catching skill with restricted vision encouraged individuals to explore additional task-relevant information, improving a full-vision task (transfer) performance (Bennett et al., 1999). We speculate that practicing passes with the demanding futsal task constraints enhanced players' ability to search for and to exploit information that specified affordances in SOC. Futsal requires a continuous (re)attunement to informational constraints that support the passing action, which might have facilitated discovery and stabilisation of functional perception and action couplings with the less demanding soccer task constraints. Conversely, practicing the passing skill in a relatively large space with more time to act did not facilitate soccer player's ability to discover functional movement solutions in FUT, and performance did not improve.

Practitioners should be encouraged to manipulate task constraints during training to expedite their players' learning and transfer of skills. The number of players and pitch dimensions (potentially the main constraints that promote transfer) are typically reduced in small-sided games during training (Davids, Araújo, Correia, & Vilar, 2013), and this study demonstrated how the use of a smaller playing area can improve passing performance in a larger playing area. Conversely, practicing within a large playing area may limit performance in a smaller playing area. Furthermore, given its more predictable trajectory and being easier to kick (Peacock et al., 2017), the futsal ball might have also played a role in developing futsal players' ability to explore and exploit the informational constraints that supported the passing action. It has been recently suggested that futsal practice can fast-track the development of soccer-related skills (Travassos et al., 2018). While being limited to short passing skill with the sample constraints (i.e. ball, playing surface and playing area), the results of this study suggest soccer practitioners to introduce futsal task constraints in their practice to improve their athletes' passing performance.

We interpreted the results as being influenced by long-term practice with domain-specific task constraints. However, we cannot exclude that factors other than task constraints contributed to the results, e.g. futsal participants potentially performed more activities that promoted skill adaptability relative to soccer participants. A future research direction would be to conduct a longitudinal study manipulating domain-specific task constraints to evaluate direct correlations between certain task constraints and skill transfer in similar domains.

Conclusions

This study highlighted how learning a passing skill with futsal task constraints (i.e. small area and short time to act) promotes transfer to soccer task constraints (i.e. larger space and longer time to act) than vice-versa. Futsal players improved their passing performance when they moved to the soccer-like task, while soccer group's passing performance remained stable. The more adaptable perceptual behaviour of the futsal group is suggested to be one of the main mechanisms promoting skill transfer. Practitioners are encouraged to reduce playing area during training to enhance players' ability to functionally couple perception and action when playing in larger playing area.

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ORCID

Derek Panchuk http://orcid.org/0000-0002-1270-561X

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